

Electromagnetic Waves: From Maxwell's Equations to Energy Transport in Matter

Every equation in this essay was verified symbolically with SymPy (1.14.0); the verification scripts are provided alongside this document.

1. Maxwell's equations in vacuum

In vacuum there are no sources — no free charge density ρ and no current density $\mathbf{J}$ — and Maxwell's equations take their simplest, fully symmetric form. In differential form:

$$\nabla \cdot \mathbf{E} = 0 \tag{1}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{2}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{3}$$

$$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{4}$$

Here $\mathbf{E}$ and $\mathbf{B}$ are the electric and magnetic fields, ε_0 is the vacuum permittivity, and μ_0 is the vacuum permeability. The four equations have a clean physical reading:

- **(1) Gauss's law for electricity (source-free).** The absence of charge means electric field lines have no beginning or end in empty space; the flux of $\mathbf{E}$ through any closed surface vanishes.
- **(2) Gauss's law for magnetism.** There are no magnetic monopoles; magnetic field lines always close on themselves.
- **(3) Faraday's law.** A time-varying magnetic field induces a circulating electric field.
- **(4) The Ampère-Maxwell law.** A time-varying electric field (the displacement current) induces a circulating magnetic field.

The last two equations are the engine of the whole theory: a changing $\mathbf{B}$ creates $\mathbf{E}$, and a changing $\mathbf{E}$ creates $\mathbf{B}$. That mutual induction is what makes self-sustaining electromagnetic waves possible.

Two constants are already hiding in these equations. Their product

$$\mu_0 \varepsilon_0 = \frac{1}{c^2}$$

defines a speed $c = 1/\sqrt{\mu_0 \varepsilon_0} \approx 3 \times 10^8$ m/s. As the next section shows, this is not a coincidence: it is the speed at which the equations force any electromagnetic disturbance to propagate.

2. The wave equation

The wave equations for $\mathbf{E}$ and $\mathbf{B}$ follow by one curl and one substitution each.

Derivation for $\mathbf{E}$. Take the curl of Faraday's law (3):

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right).$$

Apply the vector identity $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ on the left, and interchange the curl and the time derivative on the right (they commute for smooth fields):

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}).$$

Now the two source-free Gauss laws do their work. By (1), $\nabla \cdot \mathbf{E} = 0$, so the first term vanishes. Substituting the Ampère-Maxwell law (4) for $\nabla \times \mathbf{B}$:

$$-\nabla^2 \mathbf{E} = -\frac{\partial}{\partial t} \left(\mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right).$$

Since $\mu_0 \varepsilon_0$ is a constant it comes out of the derivative, and cancelling the minus signs gives

$$\boxed{\nabla^2 \mathbf{E} = \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}} \quad (5)$$

Derivation for $\mathbf{B}$. Symmetrically, take the curl of (4), use the vector identity with $\mathbf{B}$, drop the $\nabla(\nabla \cdot \mathbf{B})$ term via (2), and substitute Faraday's law (3) for $\nabla \times \mathbf{E}$:

$$-\nabla^2 \mathbf{B} = \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(-\frac{\partial \mathbf{B}}{\partial t} \right) \implies \boxed{\nabla^2 \mathbf{B} = \frac{1}{c^2} \frac{\partial^2 \mathbf{B}}{\partial t^2}} \quad (6)$$

What this means. Both fields independently satisfy the standard wave equation in three dimensions, propagating at speed

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 2.998 \times 10^8 \text{ m/s}.$$

Equations (5)–(6) are the mathematical statement that a self-sustaining disturbance in $\mathbf{E}$ induces one in $\mathbf{B}$, and vice versa, and that this coupled disturbance travels through empty space at c . The identification of this c with the measured speed of light is the content of Maxwell's great unification: light is an electromagnetic wave.

3. The plane-wave solution

Because the vacuum equations are linear with constant coefficients, the natural building blocks are monochromatic plane waves. We take the ansatz

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}, \quad \mathbf{B}(\mathbf{r}, t) = \mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)},$$

with constant (possibly complex) amplitudes $\mathbf{E}_0, \mathbf{B}_0$; the physical fields are the real parts. Each differential operator acts as a simple multiplication once the common factor $f = e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$ is stripped off:

Equation	Result	Condition
$\nabla \cdot \mathbf{E} = 0$	$i \mathbf{k} \cdot \mathbf{E}_0 = 0$	$\mathbf{k} \cdot \mathbf{E}_0 = 0$

$\nabla \cdot \mathbf{B} = 0$	$i \mathbf{k} \cdot \mathbf{B}_0 = 0$	$\mathbf{k} \cdot \mathbf{B}_0 = 0$
$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$	$i \mathbf{k} \times \mathbf{E}_0 = i\omega \mathbf{B}_0$	$\mathbf{B}_0 = \frac{\mathbf{k} \times \mathbf{E}_0}{\omega}$
$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \partial_t \mathbf{E}$	$i \mathbf{k} \times \mathbf{B}_0 = -i\mu_0 \varepsilon_0 \omega \mathbf{E}_0$	$\mathbf{k} \times \mathbf{B}_0 = -\mu_0 \varepsilon_0 \omega \mathbf{E}_0$

3.1 The dispersion relation

Substituting $\mathbf{B}_0 = \mathbf{k} \times \mathbf{E}_0 / \omega$ into the Ampère condition gives

$$\mathbf{k} \times (\mathbf{k} \times \mathbf{E}_0) = -\mu_0 \varepsilon_0 \omega^2 \mathbf{E}_0.$$

The vector triple product $\mathbf{k} \times (\mathbf{k} \times \mathbf{E}_0) = \mathbf{k}(\mathbf{k} \cdot \mathbf{E}_0) - k^2 \mathbf{E}_0$ yields

$$\mathbf{k}(\mathbf{k} \cdot \mathbf{E}_0) + (\mu_0 \varepsilon_0 \omega^2 - k^2) \mathbf{E}_0 = 0.$$

Dotting with $\mathbf{E}_0$ and using the transversality $\mathbf{k} \cdot \mathbf{E}_0 = 0$ leaves

$$(\mu_0 \varepsilon_0 \omega^2 - k^2) |\mathbf{E}_0|^2 = 0.$$

For a nontrivial wave ($\mathbf{E}_0 \neq 0$),

$$\boxed{\omega = c |\mathbf{k}|, \quad c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}.} \quad (7)$$

This is precisely the dispersion relation of the wave equation (5)-(6), confirming that the plane-wave ansatz is consistent with the PDE.

3.2 Polarization structure

All the geometric facts about plane waves follow from the single relation $\mathbf{B}_0 = \mathbf{k} \times \mathbf{E}_0 / \omega$:

- **Transversality.** $\mathbf{k} \cdot \mathbf{E}_0 = 0$ by Gauss's law. For $\mathbf{B}_0$:

$$\mathbf{k} \cdot \mathbf{B}_0 = \frac{\mathbf{k} \cdot (\mathbf{k} \times \mathbf{E}_0)}{\omega} \equiv 0,$$

a scalar triple product with a repeated vector. Gauss's law for magnetism is satisfied automatically.

- **Orthogonality of $\mathbf{E}$ and $\mathbf{B}$.**

$$\mathbf{E}_0 \cdot \mathbf{B}_0 = \frac{1}{\omega} \mathbf{E}_0 \cdot (\mathbf{k} \times \mathbf{E}_0) = 0,$$

again a repeated vector in a triple product.

- **Amplitude ratio.** Since $\mathbf{E}_0 \perp \mathbf{k}$,

$$|\mathbf{B}_0| = \frac{k|\mathbf{E}_0|}{\omega} = \frac{|\mathbf{E}_0|}{c}.$$

The three vectors $\mathbf{E}_0$, $\mathbf{B}_0$, and $\hat{\mathbf{k}}$ therefore form a mutually orthogonal, right-handed triad with $\mathbf{B}_0 = \frac{1}{c} \hat{\mathbf{k}} \times \mathbf{E}_0$. In particular $\mathbf{E} \times \mathbf{B}$ points along $\hat{\mathbf{k}}$ — the direction of propagation. (The direction of propagation of the *energy* is confirmed a posteriori by the Poynting vector in §4.)

3.3 Explicit real solutions

Writing $\varphi = \mathbf{k} \cdot \mathbf{r} - \omega t$:

$$\mathbf{E} = \text{Re}[\mathbf{E}_0 e^{i\varphi}] = \Re(\mathbf{E}_0) \cos \varphi - \Im(\mathbf{E}_0) \sin \varphi.$$

- **Linear polarization** ($\mathbf{E}_0$ real): $\mathbf{E} = \mathbf{E}_0 \cos \varphi$; the field oscillates along a fixed direction.
- **Circular polarization** (e.g. $\mathbf{E}_0 = \frac{E_0}{\sqrt{2}}(\hat{x} + i\hat{y})$ for $\mathbf{k} = k\hat{z}$):

$$\mathbf{E} = \frac{E_0}{\sqrt{2}} (\hat{x} \cos \varphi - \hat{y} \sin \varphi),$$

a vector of constant magnitude rotating at angular frequency ω .

3.4 A verified example

With $\mathbf{k} = k\hat{z}$, $\mathbf{E}_0 = E_x \hat{x}$, and $\omega = ck$, the relation $\mathbf{B}_0 = \hat{\mathbf{k}} \times \mathbf{E}_0 / c$ gives $\mathbf{B}_0 = \frac{E_x}{c} \hat{y}$. Substituting this pair back into all four Maxwell equations (done symbolically with SymPy) yields exactly zero for each:

$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \partial_t \mathbf{B} = \mathbf{0}, \quad \nabla \times \mathbf{B} - \mu_0 \varepsilon_0 \partial_t \mathbf{E} = \mathbf{0},$$

and $\mathbf{E} \times \mathbf{B} \propto +\hat{z}$, confirming propagation along $\hat{\mathbf{k}}$.

Finally, linearity implies that any superposition

$$\mathbf{E}(\mathbf{r}, t) = \int \tilde{\mathbf{E}}_0(\mathbf{k}) e^{i(\mathbf{k} \cdot \mathbf{r} - c|\mathbf{k}|t)} d^3 k$$

is also a solution — which is why arbitrary initial data can be built from plane waves.

4. Energy density and the Poynting vector in vacuum

The conservation law for electromagnetic energy is **Poynting's theorem**, derived as follows.

4.1 Derivation

From the Ampère–Maxwell law,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \implies \mathbf{J} = \frac{1}{\mu_0} \left(\nabla \times \mathbf{B} - \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right).$$

The power per unit volume delivered by the fields to the charges is $\mathbf{J} \cdot \mathbf{E}$:

$$\mathbf{J} \cdot \mathbf{E} = \frac{1}{\mu_0} (\nabla \times \mathbf{B}) \cdot \mathbf{E} - \varepsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}.$$

The key vector identity (verified term-by-term symbolically),

$$(\nabla \times \mathbf{B}) \cdot \mathbf{E} = \nabla \cdot (\mathbf{E} \times \mathbf{B}) + (\nabla \times \mathbf{E}) \cdot \mathbf{B},$$

rearranges this to

$$\mathbf{J} \cdot \mathbf{E} = \frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}) + \frac{1}{\mu_0} (\nabla \times \mathbf{E}) \cdot \mathbf{B} - \varepsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}.$$

Faraday's law, $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$, together with the elementary "half trick" $\mathbf{E} \cdot \partial_t \mathbf{E} = \frac{1}{2} \partial_t E^2$ (and its magnetic analogue) gives

$$\mathbf{J} \cdot \mathbf{E} = \frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}) - \frac{\partial}{\partial t} \left[\frac{\varepsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2 \right].$$

Rearranging into conservation-law form:

$$\boxed{\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}, \quad u = \frac{\varepsilon_0}{2} |\mathbf{E}|^2 + \frac{1}{2\mu_0} |\mathbf{B}|^2, \quad \mathbf{S} \equiv \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}. \quad (8)}$$

Reading (8). u is the electromagnetic energy density — half stored in the electric field, half in the magnetic. $\mathbf{S}$, the **Poynting vector**, is the energy flux (in W/m²). The statement: the change of field energy in a volume plus the net flux through its surface equals the energy handed to the charges inside. Integrating over a volume V ,

$$\frac{d}{dt} \int_V u dV + \oint_{\partial V} \mathbf{S} \cdot d\mathbf{a} = - \int_V \mathbf{J} \cdot \mathbf{E} dV.$$

In vacuum $\mathbf{J} = 0$, and (8) becomes a pure conservation law: $\partial_t u + \nabla \cdot \mathbf{S} = 0$, a continuity equation for electromagnetic energy with density u and current $\mathbf{S}$.

4.2 The plane wave

Take $\mathbf{E} = E_0 \cos(kz - \omega t) \hat{x}$ and $\mathbf{B} = \frac{E_0}{c} \cos(kz - \omega t) \hat{y}$. The energy density splits into two contributions that are **equal point by point**:

$$u_E = \frac{\varepsilon_0}{2} E_0^2 \cos^2 \varphi = u_B = \frac{1}{2\mu_0} \frac{E_0^2}{c^2} \cos^2 \varphi,$$

the equality following from $\varepsilon_0 = 1/(\mu_0 c^2)$. Hence

$$u = \varepsilon_0 E_0^2 \cos^2 \varphi,$$

and the Poynting vector is

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} = \frac{E_0^2}{\mu_0 c} \cos^2 \varphi \hat{z} = c u \hat{z}.$$

The energy flows along $\hat{\mathbf{k}}$ at the speed of light. Averaging over one period ($\langle \cos^2 \varphi \rangle = \frac{1}{2}$):

$$\langle u \rangle = \frac{\varepsilon_0 E_0^2}{2}, \quad \langle \mathbf{S} \rangle = \frac{E_0^2}{2\mu_0 c} \hat{z} = c \langle u \rangle \hat{z}.$$

The magnitude of the time-averaged flux is the **intensity** of the wave,

$$I \equiv \langle S \rangle = \frac{E_0^2}{2\mu_0 c} = \frac{c\varepsilon_0 E_0^2}{2}. \quad (9)$$

Numerically, $I \approx 1.33 \times 10^{-3} \text{ W/m}^2$ per $(\text{V/m})^2$ of field amplitude.

The physics in one sentence: the wave's energy is stored half in $\mathbf{E}$ and half in $\mathbf{B}$ in exact synchrony, and it travels as the flux $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$ moving at c ; in vacuum nothing is lost.

5. Maxwell's equations in matter

5.1 Setup: macroscopic fields

Inside material the microscopic fields are averaged into **macroscopic fields** $\mathbf{E}, \mathbf{B}, \mathbf{D}, \mathbf{H}$, with sources split into free ($\rho_f, \mathbf{J}_f$) and bound parts (absorbed into $\mathbf{D}$ and $\mathbf{H}$). The macroscopic equations are

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \rho_f, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, & \nabla \times \mathbf{H} &= \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}. \end{aligned}$$

The material's response enters through **constitutive relations**. The standard linear, isotropic, homogeneous medium with Ohmic conduction is

$$\mathbf{D} = \varepsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}, \quad \mathbf{J}_f = \sigma \mathbf{E},$$

where ε and μ are the permittivity and permeability of the medium and σ its conductivity ($\sigma = 0$: lossless dielectric; $\sigma \rightarrow \infty$: perfect conductor). We take the medium source-free, $\rho_f = 0$.

5.2 Energy density and Poynting vector in matter

Repeating the derivation of §4.1 with $\mathbf{H}$ and $\mathbf{D}$ in place of $\mathbf{B}/\mu_0$ and $\varepsilon_0 \mathbf{E}$, the conduction current now appears as a *sink* of field energy (Joule heating):

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J}_f \cdot \mathbf{E}, \quad u = \frac{1}{2} (\mathbf{E} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{H}), \quad \mathbf{S} = \mathbf{E} \times \mathbf{H}. \quad (10)$$

The full identity $\partial_t u + \nabla \cdot \mathbf{S} + \mathbf{J}_f \cdot \mathbf{E} - (\nabla \times \mathbf{E} + \partial_t \mathbf{B}) \cdot \mathbf{H} = 0$ holds for *generic* fields (verified symbolically), so (10) is exact on solutions. For the Ohmic law, $\mathbf{J}_f \cdot \mathbf{E} = \sigma |\mathbf{E}|^2 \geq 0$: field energy is converted to heat at that rate.

Two differences from vacuum: $u = \frac{1}{2} (\varepsilon E^2 + B^2/\mu)$ and $\mathbf{S} = \mathbf{E} \times \mathbf{H} = \mathbf{E} \times \mathbf{B}/\mu$. In a *linear* medium the factor $\frac{1}{2}$ in u is exact; nonlinear or dispersive media require a more careful stored-energy density (Brillouin's formula).

5.3 The wave equation in matter

Curl Faraday's law, apply the vector identity, and substitute Ampère's law with the constitutive relations:

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) = -\mu \frac{\partial}{\partial t} \left(\sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \right).$$

With $\nabla \cdot \mathbf{E} = 0$ (source-free) and constant ε, μ, σ :

$$\boxed{\nabla^2 \mathbf{E} = \mu\varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} + \mu\sigma \frac{\partial \mathbf{E}}{\partial t}.} \quad (11)$$

This is a **damped** wave equation: the extra first-time-derivative term $\mu\sigma \partial_t \mathbf{E}$ is the loss. For $\sigma = 0$ it reduces to the vacuum form with propagation speed $v = 1/\sqrt{\mu\varepsilon}$. An identical equation holds for $\mathbf{H}$.

5.4 Plane waves and the complex dispersion relation

Insert the plane-wave ansatz $\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)}$ with a **complex** wave vector. Stripping the exponential, the four macroscopic equations give (all verified symbolically):

- $\nabla \cdot \mathbf{D} = 0 \Rightarrow \mathbf{k} \cdot \mathbf{E}_0 = 0$ — the wave is still **transverse**;
- $\nabla \times \mathbf{E} = -\partial_t \mathbf{B} \Rightarrow \mathbf{B}_0 = \frac{\mathbf{k} \times \mathbf{E}_0}{\omega}$ — unchanged in form;
- $\nabla \cdot \mathbf{B} = 0$ — automatic (triple product with repeated vector);
- $\nabla \times \mathbf{H} = \mathbf{J}_f + \partial_t \mathbf{D} \Rightarrow$ the **dispersion relation**

$$\boxed{\mathbf{k}^2 = \mu\varepsilon\omega^2 + i\mu\sigma\omega.} \quad (12)$$

Equation (12) is the entire story of propagation in a linear conducting medium. Writing $\mathbf{k} = \kappa \hat{\mathbf{k}}$ with $\kappa = \beta + i\alpha$:

$$\beta^2 - \alpha^2 = \mu\varepsilon\omega^2, \quad 2\alpha\beta = \mu\sigma\omega,$$

where β is the **phase** wavenumber (wavelength, phase velocity $v_p = \omega/\beta$) and α is the **attenuation** constant: the field decays as $e^{-\alpha z}$. The **skin depth** is $\delta = 1/\alpha$, and the **intrinsic impedance** of the medium,

$$\eta = \frac{\omega\mu}{\kappa} = \sqrt{\frac{\mu}{\varepsilon - i\sigma/\omega}},$$

is complex when $\sigma \neq 0$, encoding both the amplitude ratio $|\mathbf{H}_0| = |\kappa||\mathbf{E}_0|/\omega$ and a phase lag between $\mathbf{E}$ and $\mathbf{H}$.

Good-conductor limit. When conduction current dominates displacement current ($\sigma \gg \omega\varepsilon$):

$$\kappa \approx (1+i)\sqrt{\frac{\mu\sigma\omega}{2}}, \quad \delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\mu\sigma\omega}}, \quad \eta \approx \sqrt{\frac{\omega\mu}{\sigma}} e^{-i\pi/4}.$$

The $\pi/4$ phase means $\mathbf{H}$ lags $\mathbf{E}$ by 45° inside a good conductor.

5.5 Worked examples (numerically verified)

Lossless dielectric — “glass” with $n = 2$. Take $\varepsilon = 4\varepsilon_0$, $\mu = \mu_0$. Then $v = c/2$, $n = c\sqrt{\mu\varepsilon} = 2$, and $\eta = \eta_0/2$. For $\mathbf{E} = E_0 \cos(kz - \omega t)\hat{x}$, $\mathbf{B} = \frac{E_0}{v} \cos(kz - \omega t)\hat{y}$, all four macroscopic Maxwell equations evaluate to exactly zero, and the energy results are

$$u(t) = \varepsilon E_0^2 \cos^2 \varphi, \quad \mathbf{S} = \frac{E_0^2}{\mu v} \cos^2 \varphi \hat{z} = v u \hat{z},$$

$$\langle u \rangle = \frac{1}{2} \varepsilon E_0^2, \quad \langle \mathbf{S} \rangle = \frac{E_0^2}{2\eta} \hat{z} = \frac{1}{2} \varepsilon v E_0^2 \hat{z}.$$

Compared with vacuum: the speed drops to c/n , the impedance to η_0/n , and the energy flux for a given field amplitude drops by $1/n^2$.

Lossy conductor — copper at 60 Hz. With $\sigma = 5.8 \times 10^7$ S/m, $\mu = \mu_0$, $\varepsilon = \varepsilon_0$, $f = 60$ Hz, solving (12) gives

$$\beta = \alpha \approx 117.2 \text{ m}^{-1}, \quad \delta = 1/\alpha \approx 8.53 \text{ mm}, \quad |\eta| \approx 2.86 \times 10^{-6} \Omega, \quad \angle \eta = -45^\circ.$$

The good-conductor approximation $\delta = \sqrt{2/(\mu\sigma\omega)}$ reproduces the exact value to four significant figures, as expected since the displacement-to-conduction current ratio is $\omega\varepsilon_0/\sigma \approx 5.8 \times 10^{-17}$. Physically: a field penetrating copper loses a factor e every 8.5 mm, and the surface impedance is only a few micro-ohms — essentially a short circuit, which is why RF currents are confined to a skin only microns thick.

5.6 Energy balance: the dispersion relation is energy conservation

For the attenuated wave $|\tilde{\mathbf{E}}|^2 e^{-2\alpha z}$, the averaged flux and Joule loss are

$$\langle S_z \rangle = \frac{\beta}{2\omega\mu} |\tilde{E}|^2 e^{-2\alpha z}, \quad \langle \mathbf{J}_f \cdot \mathbf{E} \rangle = \frac{\sigma}{2} |\tilde{E}|^2 e^{-2\alpha z}.$$

Symbolic differentiation shows

$$-\frac{d\langle S_z \rangle}{dz} = \langle \mathbf{J}_f \cdot \mathbf{E} \rangle \iff 2\alpha\beta = \mu\sigma\omega,$$

which is exactly the imaginary part of the dispersion relation (12). The attenuation of the wave is *precisely* accounted for by the Joule heating it deposits — the two statements are the same fact, one from the wave equation, the other from Poynting's theorem. The time-averaged stored energy generalizes to

$$\langle u \rangle = \frac{1}{4} |\tilde{E}|^2 \left(\varepsilon + \frac{\beta^2 + \alpha^2}{\omega^2 \mu} \right) e^{-2\alpha z} \xrightarrow{\sigma \rightarrow 0} \frac{1}{2} \varepsilon |\tilde{E}|^2 e^{-2\alpha z}.$$

6. What changes when matter enters

Quantity	Vacuum	Linear medium
Constitutive relations	$\mathbf{D} = \varepsilon_0 \mathbf{E}, \mathbf{B} = \mu_0 \mathbf{H}$	$\mathbf{D} = \varepsilon \mathbf{E}, \mathbf{B} = \mu \mathbf{H}, \mathbf{J}_f = \sigma \mathbf{E}$
Wave equation	$\nabla^2 \mathbf{E} = \mu_0 \varepsilon_0 \ddot{\mathbf{E}}$	$\nabla^2 \mathbf{E} = \mu \varepsilon \ddot{\mathbf{E}} + \mu \sigma \dot{\mathbf{E}} (\text{damped})$
Dispersion	$k^2 = \mu_0 \varepsilon_0 \omega^2$	$k^2 = \mu \varepsilon \omega^2 + i \mu \sigma \omega (\text{complex } k)$
Speed / index	c	$v = 1/\sqrt{\mu \varepsilon}, n = c\sqrt{\mu \varepsilon} (\text{lossless})$
Intrinsic impedance	$\eta_0 \approx 377 \Omega$	$\eta = \sqrt{\mu/(\varepsilon - i\sigma/\omega)}$
Energy density	$\frac{1}{2}(\varepsilon_0 E^2 + B^2/\mu_0)$	$\frac{1}{2}(\mathbf{E} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{H})$
Poynting vector	$\mathbf{E} \times \mathbf{B}/\mu_0$	$\mathbf{E} \times \mathbf{H}$
Energy balance	$\partial_t u + \nabla \cdot \mathbf{S} = 0$	$\partial_t u + \nabla \cdot \mathbf{S} = -\sigma \mathbf{E} ^2$
Attenuation	none	$\text{skindepth} \delta = 1/\alpha; \text{good conductor } \delta = \sqrt{2/(\mu\sigma\omega)}$

The **transverse** structure, the relation $\mathbf{B}_0 = \mathbf{k} \times \mathbf{E}_0/\omega$, and the right-handed $\{\mathbf{E}, \mathbf{H}, \hat{\mathbf{k}}\}$ triad survive unchanged. Matter only changes the *magnitude* of the field ratio (via η), the *speed* (via n), and adds *attenuation* (via α) and a *phase lag* between $\mathbf{E}$ and $\mathbf{H}$ (via complex η).

Scope and caveats

This treatment covers **linear, isotropic, homogeneous, nondispersive** media. Real materials are dispersive — ε and σ depend on ω — which makes u and $\mathbf{S}$ frequency-dependent and calls for the Brillouin energy formula. Anisotropic media (crystals, waveguides) have $\mathbf{D} = \varepsilon_{ij}E_j$, and the waves become elliptically polarized with $\mathbf{E}$ no longer strictly transverse to $\mathbf{k}$. For the standard introductory treatment, however, the picture above is complete.

Appendix: Verification

All algebraic claims in this essay were checked symbolically with SymPy 1.14.0. The scripts:

- `maxwell_plane_wave.py` — plane-wave conditions, dispersion relation, polarization identities, and the explicit $k\hat{z}$ example (all four Maxwell equations evaluate to 0).
- `poynting_derivation.py` — the identity $\mathbf{J} \cdot \mathbf{E} + \partial_t u + \nabla \cdot (\mathbf{E} \times \mathbf{B})/\mu_0 = (\nabla \times \mathbf{E} + \partial_t \mathbf{B}) \cdot \mathbf{B}/\mu_0$ for generic fields; plane-wave u , $\mathbf{S}$, $\partial_t u + \nabla \cdot \mathbf{S} = 0$, $\mathbf{S} = cu\hat{z}$, and cycle averages.
- `maxwell_matter.py` — Poynting's theorem in matter for generic fields; the lossy dispersion relation $k^2 = \mu\varepsilon\omega^2 + i\mu\sigma\omega$; the damped wave equation; the $n = 2$ lossless example (all four macroscopic equations = 0, $\langle \mathbf{S} \rangle = v\langle u \rangle \hat{z} = E_0^2/(2\eta)$); copper at 60 Hz ($\beta, \alpha, \delta, \eta_c$); and the energy-balance identity $-d\langle S_z \rangle/dz = \langle \mathbf{J}_f \cdot \mathbf{E} \rangle \Leftrightarrow 2\alpha\beta = \mu\sigma\omega$.